Further leaves and elementary halving criteria

Human Compression project: N014 continuation

We use the complete parity-cover theorem and its notation. Thus Γ is a residually connected geometry with a simple vertex–edge graph Q on types $0, 1$, satisfying (B2): an edge is incident with a higher object exactly when both its endpoints are. For a higher flag F, Q_F is its common vertex–edge shadow. It is connected and contains an edge. Write $D(Q)$ for the bipartite double cover, and choose a component C. The halved geometry has the vertices of C as its two new types; a higher object is (x, C_x), where C_x is a component of $D(Q_x)$ inside C. Incidence is membership for a lifted vertex and nonempty component intersection for incident higher objects. The parity-cover theorem proves flag extension without an additional parity hypothesis. It proves residual connectedness precisely under parity compatibility. Those full proofs are retained inputs, not inferred from this note's finite example.

Two further elementary consequences recover the graph and duality clauses of the source. If Q has girth g, the girth of its double cover is the smaller of the least even cycle length in Q and twice the least odd cycle length, omitting a term when that kind of cycle does not exist. Indeed an even cycle lifts to a cycle of its own length, and an odd cycle traversed twice lifts to a cycle of twice its length. Conversely a shortest cover cycle projects to a closed walk without immediate reversals. If the projection is simple, it is an even cycle. If it is not simple, cutting at repeated vertices either exhibits an even cycle no longer than the walk, or exhibits at least two odd cycles, counted with multiplicity. The connecting paths contribute only additional length. This gives the claimed lower bound. Thus, if g is even, the new gonality is $g/2$; if g is odd, it is $\min\{g, k/2\}$, where k is the least even cycle length when one exists. This is source Lemma 3.5.

When Q is nonbipartite, $C = D(Q)$ and the involution $(v, \epsilon) \mapsto (v, 1 - \epsilon)$ exchanges the new types. It sends a higher component to its image under the same involution, preserving all incidences. It is the duality of Proposition 3.15.

1 The second leaf

Suppose that two further types k, l also satisfy (B1) and (B2), with type k as vertices. Write the two endpoints of a type-l object y as x_1, x_2. Conditions (B2) at this leaf give

$$Q_y = Q_{x_1} \cap Q_{x_2}$$

as vertex–edge subgraphs, even though $\{x_1, x_2\}$ is not a flag. In particular Q_y is contained in each endpoint shadow.

Theorem 1. *The halved geometry satisfies (B1) and (B2) at (k, l) if and only if every y with bipartite Q_y has at least one endpoint x_j with bipartite Q_{x_j}.*

Proof. Fix a lift (y, C_y). Since C_y is connected and is a subgraph of each $D(Q_{x_j})$, it lies in a unique component C_{x_j}. These are exactly its two lifted endpoints. An endpoint component meeting C_y must be that component; hence there are always exactly two endpoints. Two distinct lifted edges with the same endpoints must project to the same y, by original (B1). Such a pair exists exactly when Q_y is bipartite, giving two components, while both Q_{x_j} are nonbipartite, so both components are contained in the same endpoint lifts. If an endpoint shadow is bipartite, its two components separate the two lifts of y. This proves that the displayed condition is equivalent to (B1).

Assume the condition, and consider (B2). Any object incident with (y, C_y) is incident with both its endpoints: for a lifted vertex this follows from $C_y \subseteq C_{x_j}$; for a higher object it follows from the same inclusion and original (B2).

Conversely, let an object be incident with both endpoints. First let it be a lifted vertex v^ϵ. Original (B2) puts v in Q_y. If Q_y is nonbipartite, its only component contains both lifts of every vertex, so $v^\epsilon \in C_y$. Otherwise choose an endpoint with bipartite shadow. The restriction of its selected component to $D(Q_y)$ is exactly C_y, so the same conclusion follows.

Now let the object be (z, C_z) of a further type. Original (B2) gives $z * y$, so $Q_{y,z}$ is connected and contains an edge. If Q_y is nonbipartite, $C_y = D(Q_y)$; since C_z projects onto every vertex of Q_z, it meets C_y over a vertex of $Q_{y,z}$. Otherwise choose a bipartite endpoint shadow Q_x. The selected components C_z, C_x meet by the given incidence. Their restrictions to the connected graph $Q_{z,x}$ are compatible: C_x selects one bipartite-cover component, and C_z either selects that component or contains both. Consequently their intersection projects onto every vertex of $Q_{z,x}$. Restrict to $Q_{y,z} \subseteq Q_{z,x}$. The component selected by C_x there is the restriction of C_y, so C_z meets C_y. This proves the converse incidence, and hence (B2). □

If an automorphism preserving the types exchanges x_1, x_2 while fixing y, their shadows have the same bipartiteness. In particular this holds in a flag-transitive original geometry. In that case the criterion reduces to the one printed in Proposition 3.17 of Piedade–Tranchida: along every incidence $x * y$, either both shadows are bipartite or neither is. In general it suffices that one endpoint is bipartite. The necessity argument in the printed proof implicitly identifies the bipartiteness of the two endpoint shadows.

A finite example distinguishing the two criteria

Let $V = \mathbb{Z}/5\mathbb{Z}$, and let Q be its cycle, with edges $e_i = \{i, i+1\}$. Use four types. Types $0, 1$ are $V, E(Q)$. Type 2 consists of X and x_j for $j \in V$; type 3 consists of y_j for $j \in V$. The vertex shadow of X is V, and those of x_j, y_j are $V \setminus \{j\}$. A vertex or edge is incident with one of these objects exactly when it is contained in its shadow. The only incidences between types $2, 3$ are

$$X * y_j, \qquad x_j * y_j.$$

The two truncations $(0, 1)$ and $(2, 3)$ are respectively the cycle and a star with five edges. They satisfy (B1). They also satisfy (B2): the induced subgraph on $V \setminus \{j\}$ is the path $Q - j$, and the intersection of the shadows of X, x_j is that same path.

Every flag extends to a chamber. If a type-3 object, or a type-2 object other than X, is specified, the remaining vertex and edge can be chosen in the nonempty path $Q - j$; a specified vertex has a neighbor in that path. Complete the higher types with y_j and either X or x_j, respecting any prescribed object. If neither is specified, choose any compatible cycle edge and then any j outside its two endpoints. These choices also cover flags already containing an edge.

Here is a complete check of residual connectedness. For a flag of types $0, 1$, the remaining incidence graph is the star at X with its incident edges subdivided by the available y_j. For a flag (v, X), the two incident cycle edges are joined through any y_j with $j \notin \{v-1, v, v+1\}$; every other available y_j is incident with at least one of the two edges. For (v, x_j) the remaining graph is a star at y_j. Flags of types $0, 3$ give $K_{d,2}$ with $d = 1$ or 2; flags of types $1, 2$ give $K_{2,3}$ or $K_{2,1}$; flags of types $1, 3$ give $K_{2,2}$. Finally a flag of types $2, 3$ has the vertex–edge incidence graph of $Q - j$ as its residue. These are all six possible pairs of types.

In a residue at a vertex v, the object X joins all remaining edges, and every remaining x_j, y_j is incident with one of them. In a residue at an edge its two endpoints and X form a connected core to which every remaining object attaches. Residues at X, x_j or y_j have the connected vertex–edge core Q or $Q - j$, and the other objects attach to it. The whole geometry has the core Q. Thus all residues of rank at least two are connected.

Now $Q_X = C_5$ is nonbipartite, whereas every $Q_{x_j} = Q_{y_j}$ is the bipartite path on four vertices. The printed condition fails at each $X * y_j$, but Theorem 1 applies using endpoint x_j. The halved geometry therefore still satisfies both conditions at $(2, 3)$. It is also residually connected: every higher flag other than $\{X\}$ has a bipartite constituent, so the parity condition is automatic. The object counts before and after halving are $(5, 5, 6, 5)$ and $(5, 5, 11, 10)$, respectively. Both geometries have 60 chambers. A separate literal incidence program checks all 322 original and 362 transformed flags. The preceding proof does not rely on that program. The example is not thin or regular; it addresses precisely the general assertion of Proposition 3.17 and the corresponding clause of Theorem 3.18.

2 Stabilizers and the induced neighbor graph

Let Γ now be a regular leaf hypertope, with distinguished involutions a, b, r_i, where a commutes with every r_i. Fix a base chamber, its vertex p, its edge e, and its other endpoint $q = ap$. Put

$$P = G_p = \langle b, r_i \rangle, \qquad R = G_{p,e} = \langle r_i \rangle.$$

All groups and actions below are the actual faithful geometric actions.

Proposition 2. *Condition (B1) holds exactly when $P \cap aPa = R$. If it fails, the marked halving subgroup $\langle aba, b, r_i \rangle$ fails the intersection property.*

Proof. The pointwise stabilizer of p, q is $P \cap aPa$. It acts transitively on the edges joining them. Indeed flag transitivity sends (p, e) to (p, e'); its element fixes p and sends the other endpoint q to itself. The stabilizer of the base edge in this action is R. Thus the set of edges with these endpoints is in bijection with $(P \cap aPa)/R$. Every edge has two distinct endpoints by thinness and the leaf condition, so (B1) is equivalent to this index being one. All pairs of adjacent vertices are equivalent under the group, so the base pair suffices.

For the new marking its two endpoint omission subgroups are $H_0 = P$ and $H_1 = aPa$, while $H_{0,1} = R$. A strict intersection therefore violates the marked intersection property. □

For a base object x_i of a higher type put $K_i = G_{x_i}$ and $P_i = P \cap K_i$. Identify vertices with left cosets gP, so the neighbors of $p = P$ form PaP/P. This convention is fixed throughout the next proposition.

Proposition 3. *Assume (B1). Condition (B2) is equivalent to*

$$K_i P \cap PaP = P_i aP \qquad (i \neq 0, 1).$$

The literal sufficient condition in source Lemma 4.6,

$$Pa \cap wP = \varnothing \quad (w \in K_i,\ wP \neq aP),$$

implies (B2), but is in fact impossible for a regular leaf hypertope of rank at least three satisfying (B1).

Proof. The vertices incident with x_i are $K_i P/P$. Since $a \in K_i$, the edges at p incident with x_i have their other endpoints $P_i aP/P$: the group P_i is transitive on the appropriate vertex–edge–object flags. Original (B1) identifies an edge uniquely with its endpoints. The displayed equality therefore says exactly that every global neighbor of p in the vertex shadow of x_i is a neighbor within its residue. This is (B2) at p, x_i; flag transitivity gives it at every vertex–object incidence. The forward implication of (B2) also follows directly from the leaf digons.

The printed condition forces any coset in the left-hand set to be aP: an intersection with PaP can be translated on the right within P to give an intersection with Pa. Hence it is a sufficient, stronger condition. To see why it cannot hold here, take $w = ba \in K_i$, which is

allowed for any higher type i. Then $ba \in Pa \cap baP$. Moreover $baP \neq aP$. Otherwise b would fix both p, q, and (B1) would force $b \in R$ by Proposition 2, contrary to the original intersection property. Thus this w contradicts the printed condition. The coset equality above gives the nonvacuous criterion. □

3 The two elementary examples

The regular polytope of type $\{3, 3, 2\}$ has group $S_4 \times C_2$, with the three adjacent transpositions on four letters and an independent central involution as its generators. Its vertices, edges and triangles are those of a tetrahedron, and it has two distinct facets, each incident with all those faces. The intersection of the two facets has several maximal triangles and no greatest proper face, so its face poset is not a lattice. Nevertheless the vertex–edge graph is K_4, and incidence with a triangle or either facet is exactly containment of the endpoints. Hence (B1) and (B2) hold. Higher flags are chains, whose common vertex shadow is the smallest shadow, so parity compatibility holds and the repaired halving theorem applies. This proves the assertion of source Example 4.8 without assuming that its face poset is a lattice.

For Example 4.9, realize the hemicube by identifying antipodal faces of the ordinary cube. It has four vertices and six edges, with graph K_4, so (B1) holds. Each of its three faces is a four-cycle using all four vertices but only four of the six edges. An omitted edge has both endpoints on that face, proving failure of (B2). Dually, each pair of its three faces has two common edges, so the opposite leaf fails (B1).

The automorphism group is the signed permutation group in dimension three modulo its central sign change, of order 24, acting faithfully as S_4 on the four antipodal vertex pairs. Choose the usual markers $a = z_1, b = s_{12}, c = s_{23}$: a acts as a double transposition and b, c as transpositions, with orders $|ab| = 4$, $|bc| = 3$ and $[a, c] = 1$. More explicitly, choose the representatives

$$A = (1, 1, 1),\ B = (1, -1, -1),\ C = (-1, 1, -1),\ D = (-1, -1, 1)$$

of the four antipodal pairs. Then $a = (AB)(CD)$, $b = (BC)$, $c = (CD)$ and $aba = (AD)$. The new markers aba, c, b are transpositions. Their successive products have order three and the two endpoints commute. The subgroup $\langle b, c\rangle$ is S_3 fixing one vertex, while aba joins that vertex to another. They are therefore the adjacent transpositions of a path on four vertices and generate S_4 with its simplex marking. Their coset geometry is the tetrahedron. This is an algebraic halving statement; it does not apply the geometric theorem after its (B2) hypothesis has failed.

4 Attribution and limits

The definitions and targets are those of Piedade–Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1, especially Propositions 3.17, 4.7, Lemmas 4.5–4.6 and Examples 4.8–4.9. The stabilizer result and examples are their conclusions. The endpoint criterion and the explicit coset equality provide the corrections and reusable arguments supplied here. The fixed preprint's general second-leaf necessity assertion is refuted; its regular flag-transitive specialization remains valid. These are source corrections, not solutions of named open conjectures. No statement is made about an uninspected journal revision, publication priority or Lean verification. A distinct mathematical review is still required.